AI-Guided Alloy Discovery

From textbook BO to a working alloy lab — and how I drive the work

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UChicago MENG 25640/35640 | April 29, 2026

- Arc A: BO → a real alloy lab
- Arc B: how I actually do this work

I sit at the seam between the BO algorithm and the powder atomizer

- Sterling Baird, Asst. Prof., BYU Mech. Eng.
- Digital Alloy Lab: AI + autonomous metallurgy
- Background: GB engineering, self-driving labs
- Co-PI on FLAIME (DOE Genesis Mission, Topic 3-B)
- Tools-on-the-bench person, not a pure theorist

The next 80 minutes are two arcs: BO-in-a-real-lab, then how I actually work

- Arc A (≈45 min): BO meets a real alloy
- Arc B (≈30 min): my research process
- Q&A throughout — please interrupt
- Slides + figures will be shared
- Goal: leave with one usable mental model

A new structural alloy still takes ~20 years and ~$100M to reach service

- Compositions are combinatorial — 15+ elements
- Process matters as much as composition
- Each experiment: hours to days, $$$
- Properties trade off — strength vs. printability
- A career = ~10^2 real samples, not 10^6

A self-driving lab is just a BO loop with hardware bolted onto every arrow

- Propose candidate (model + acquisition)
- Simulate / make the sample
- Measure properties
- Update the model → repeat
- Same shape as the BO loop you just learned

You just learned the heart of this loop: surrogate + acquisition + update

- Surrogate (usually a GP) over expensive f(x)
- Acquisition balances explore vs. exploit
- Pick the next x, run it, refit, repeat
- Built for few, expensive evaluations
- That's our regime exactly

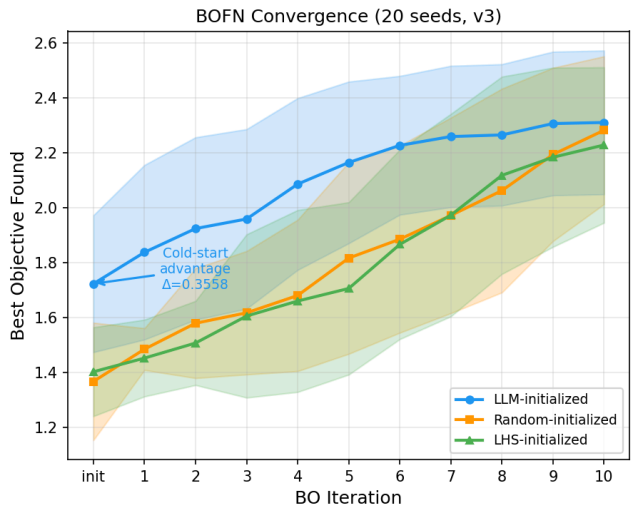
A real alloy problem violates almost every textbook BO assumption

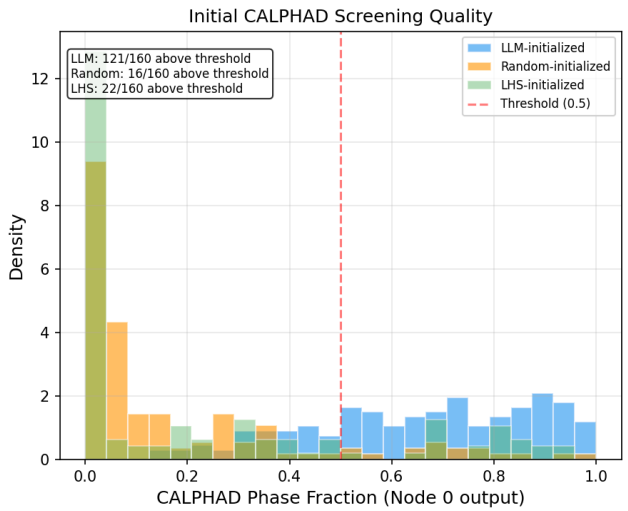
- Inputs are mixed and constrained
- Simulators have many fidelities
- Evaluations have wildly different costs
- The lab is a human + machine system
- We'll take these one at a time

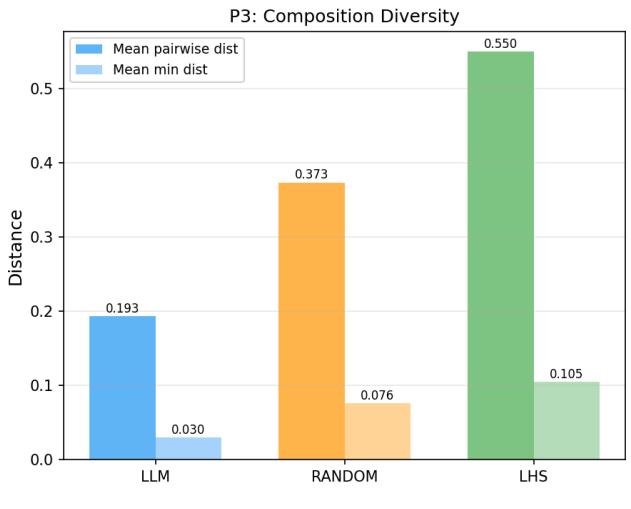
Reality 1 — the search space is mixed-categorical, continuous, and ~20-dim

- Al base + 15 alloying elements
- Compositions sum to 1 (simplex)
- Plus process params (atomization, LPBF)
- Naive grid: astronomical
- Need structure-aware BO

In high dimensions, vanilla GPs collapse — SAASBO and TuRBO are the fixes

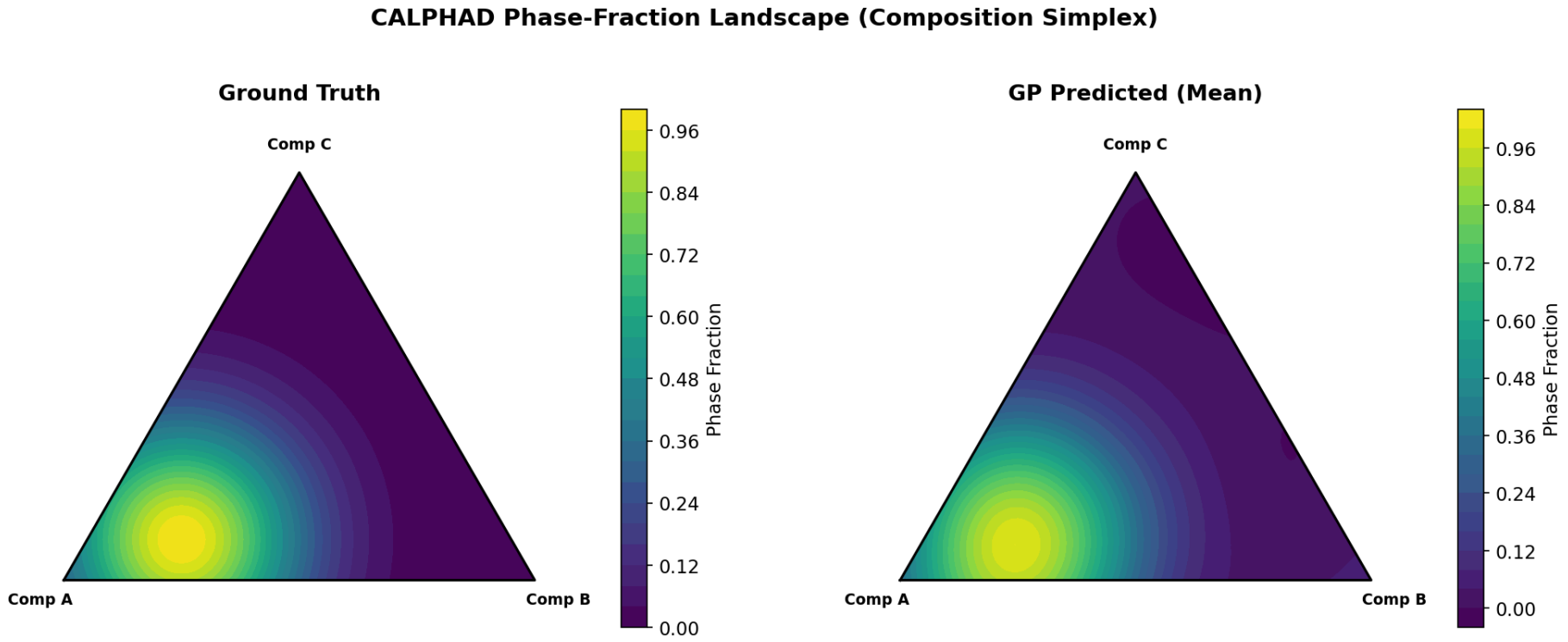






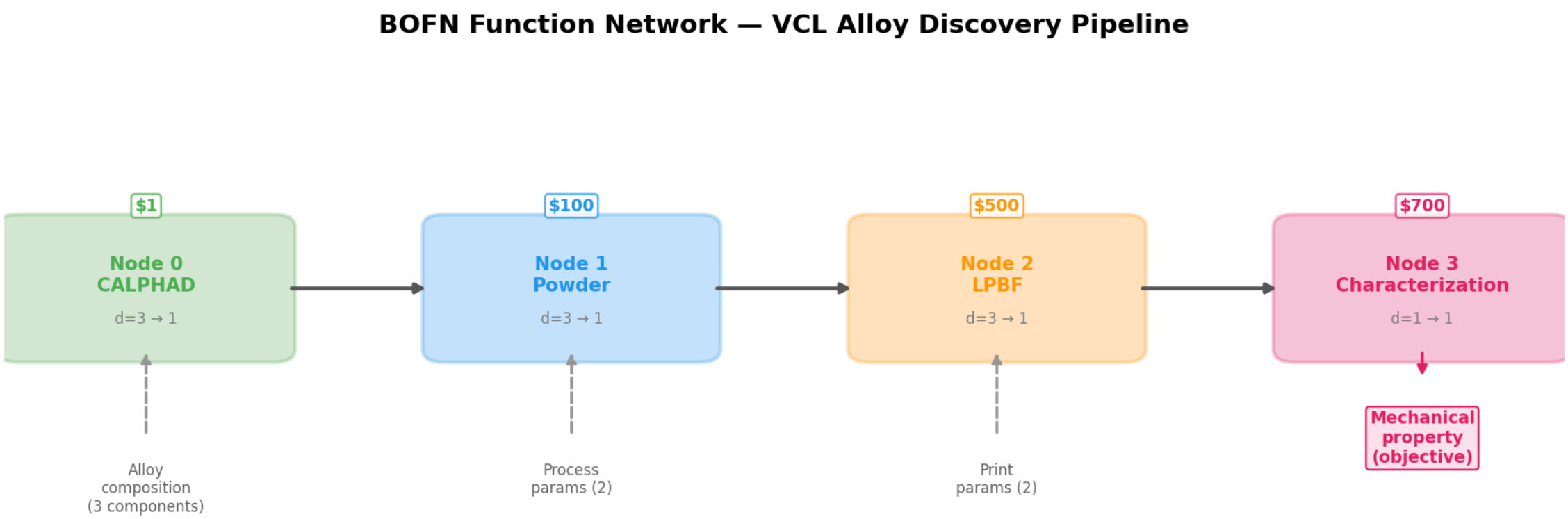
- Trust regions (TuRBO-style local models)
- Composition embeddings as features
- LLM warm-starts from literature
- GA inner solver for acquisition
- Beats random in <50 evals on benchmark

Reality 2 — your "function" is a network of cheap-to-expensive simulators



- pycalphad: cents, rough
- Thermo-Calc TCAL10 / TC-Python: $, accurate
- TC-AM Module: $$$, process-aware

BOFN treats the pipeline as a DAG and gives every node its own GP



- A function network: each node f_k is a function of its parents' outputs and free inputs
- Standard BO models the final scalar output only; BOFN models every node
- Same Bayesian machinery you know — just expressed over a graph

The DAG carries information that scalar BO throws away

- A cheap upstream node tells you a candidate is hopeless before you pay for downstream
- Per-node residuals localize WHERE the model is wrong, not just THAT it's wrong
- Mixing simulators and experiments in one framework requires this structure

The cost: more modeling work upfront. Each node needs a kernel choice and training data.

p-KGFN extends BOFN with cost-aware partial evaluations

- Knowledge gradient (KG): expected improvement in the optimum after one observation
- KGFN: KG, but observing one NODE at a time
- p-KGFN — "partial KGFN" — picks the (node, input) pair maximizing KG-per-dollar

So instead of "evaluate the whole pipeline at point x," the acquisition asks: "which single node is most worth touching next, and at what input?"

Buathong, Wan, Astudillo, Daulton, Balandat & Frazier — *ICML* (2024)

Fast-pKGFN gets a 16× speedup by sharing one Monte Carlo across nodes

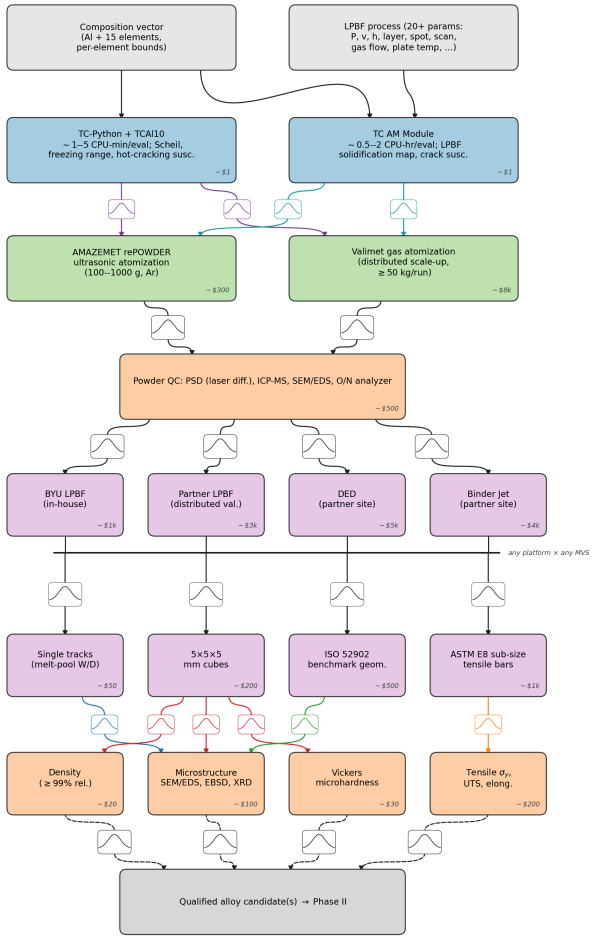
- Original p-KGFN: one nested MC per candidate node — expensive
- Fast-pKGFN: generate node-specific candidates from one global MC simulation
- Modest loss in query efficiency, up to 16× wall-clock speedup

This is the version we actually run in FLAIME.

Buathong & Frazier — *arXiv* (2025)

Our FLAIME pipeline is a DAG from CALPHAD all the way to a printed coupon

- Composition → CALPHAD → atomization → LPBF → properties
- Each box has its own surrogate
- Some edges are simulators, some are experiments
- BOFN picks where to spend next
- This is the actual production pipeline



Reality 3 — costs span six orders of magnitude across the pipeline

- Sim: seconds to minutes, dollars
- Atomization: 3.5 hours per composition
- LPBF print + heat-treat: days
- ICP-MS, SEM, tensile: hours per sample
- Acquisition must respect this

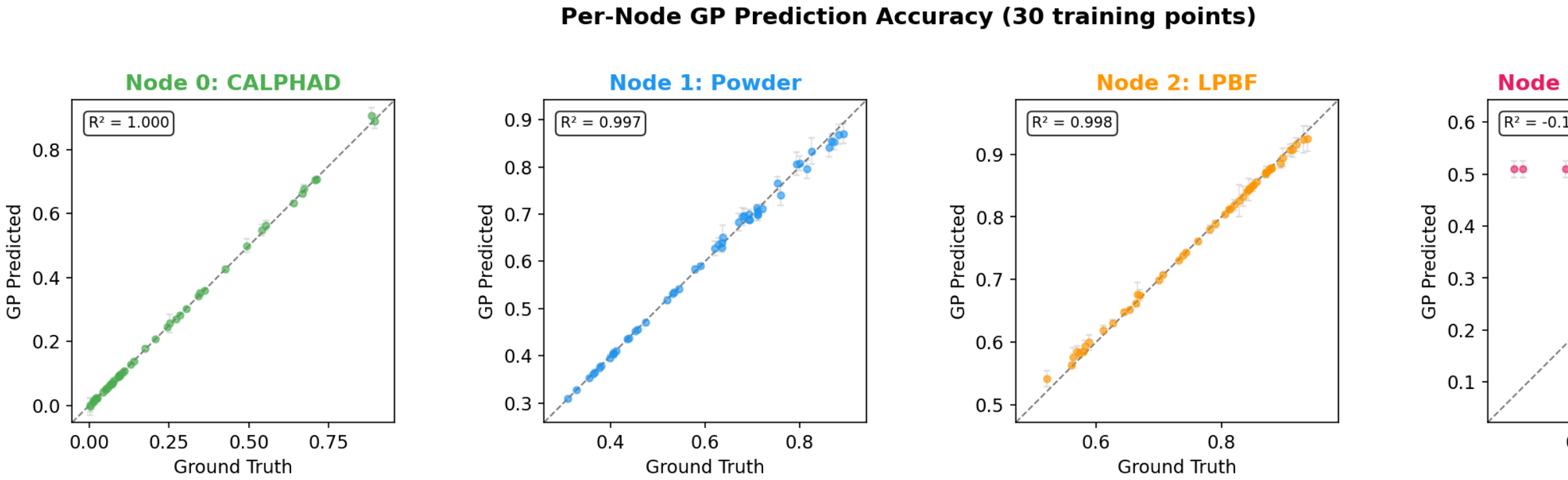
p-KGFN picks the (node, input) pair with the best information-per-dollar

[schematic to add: pKGFN choosing the cheap CALPHAD node first, deferring the expensive print node — clean single-panel diagram]

- Use EI-per-cost or knowledge-gradient-per-cost
- Cheap nodes screen; expensive nodes confirm
- Budget tracker across the whole DAG
- 55% atomizer utilization is the current bottleneck

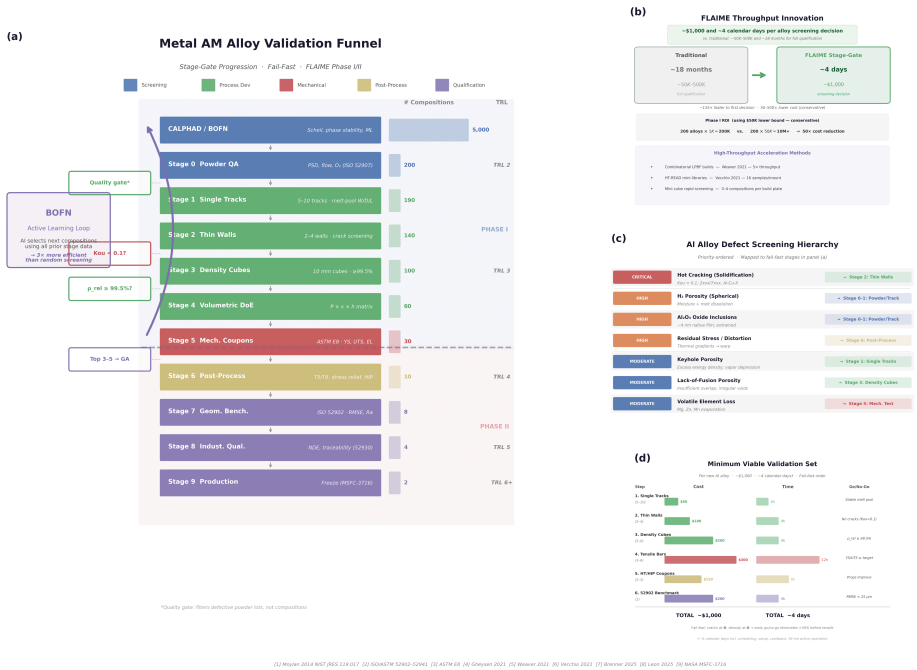
Buathong et al. *ICML* (2024) · Buathong & Frazier *arXiv* (2025) — 16× speedup

On the manufacturing benchmark, p-KGFN beats EI by 3–10× in cost-to-target



- Surrogate calibration matters more than raw R^2
- BOFN beats scalar BO in our test problem
- LLM warm-start beats cold start early

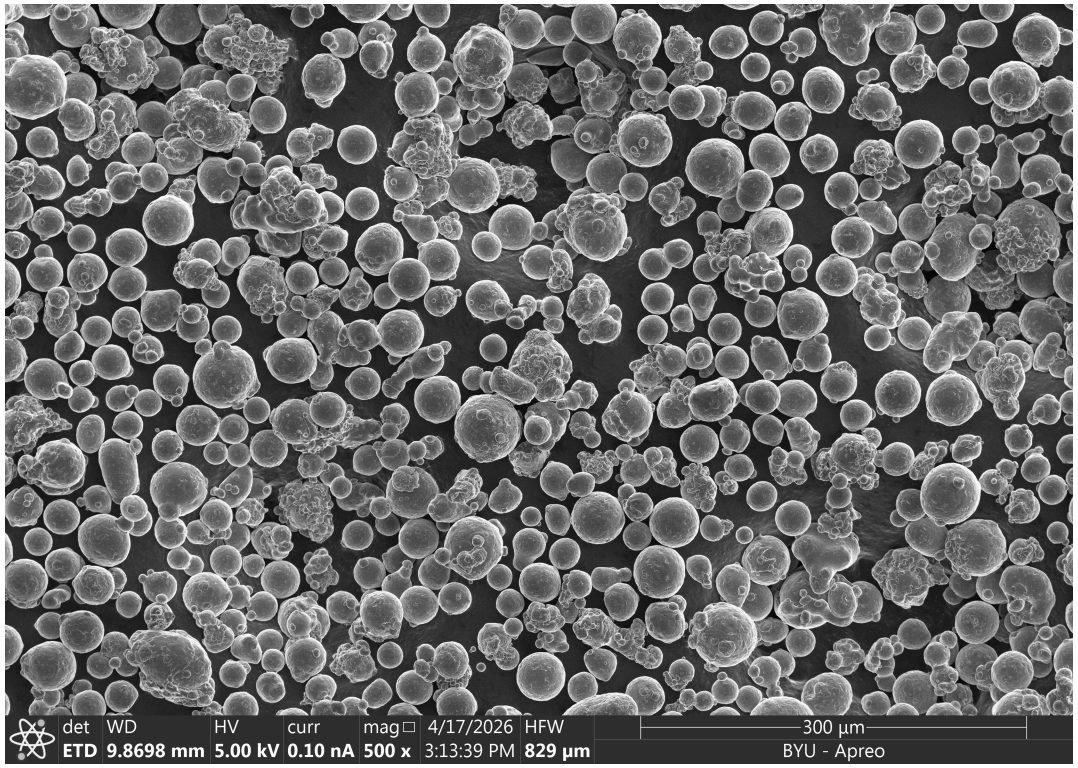
Reality 4 — the lab is a human system on a 5-day calendar



[1] Moylan 2014 NIST/IRSS 219.017 [2] ISO/ASTM 52902-52941 [3] ASTM E8 [4] Grayson 2021 [5] Weaver 2021 [6] Vecchio 2021 [7] Brenner 2025 [8] Leon 2025 [9] NASA MSFC-3716

- Operators have shifts
- Instruments have queues and downtime
- Powder has shelf life, Ar bottles run out
- Safety reviews gate new chemistries
- BO must propose plans, not points

Our hardware: induction-crucible atomizer → LPBF → SEM/EDS/ICP-MS

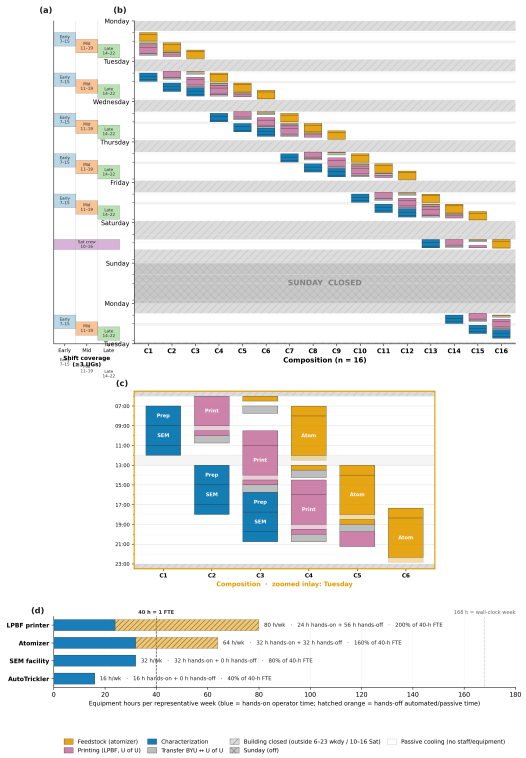


- AMAZEMET rePOWDER — induction crucible atomizer
- SLM 280 — laser powder bed fusion
- Keyence VHX-7000 — optical microscopy

The induction crucible — not wire — is what gives us full compositional freedom

- Powder hygiene is everything (oxide pickup)
- Argon handling — gloveboxes, dewpoint logs
- Crucible cross-contamination between heats
- 3.5 h per composition + cleanup
- ICP-MS confirms what you actually made

A cost-blind acquisition will tell you to do things the calendar can't allow



- Ar deliveries, operator shifts, shared XRD
- Atomization runs batched by element family
- LPBF plate = 4–8 candidates per print

"Autonomous" in a metals lab does NOT mean "self-driving" — it means scheduled

- Targets: beat Al-Mn-Cr-Zr and Al-Zr baselines
- Strength + printability + cost
- Foundational loops = reusable beyond Al
- Honest about what's unsolved
- Now: how the *research process* runs

Arc B — how I actually drive this work, day to day

- The process is half the science
- Mostly GitHub + Copilot + Edison
- Not magic, just discipline + composition
- Honest about what works and what doesn't
- Goal: things I wish someone told me

GitHub, Copilot, and Edison are each useful alone — composition is the unlock

- GitHub — versioning, review, single source of truth
- Copilot (CLI + cloud agents) — read repo, edit, commit
- Edison — literature-grounded LLM critic
- All three are useful alone
- The compounding comes from composition

Copilot + my repo + Edison's literature critique = a junior collaborator who knows the project

- Copilot reads the whole repo as context
- Copilot calls Edison as an adversary
- GitHub PRs are the audit log
- Example: PR #14 — BOFN demo + lit review + roadmap, one prompt
- Tools alone < tools composed

One PR, 100 review threads, 383 commits — collapsed into a 5-page narrative

- 383 commits, 137k additions, 373 files
- 100 review threads, 75 outdated by merge
- All driving toward 5 pages of Field 8
- Agent-assisted edits, PI in the loop
- Repo is the audit log

The single highest-leverage habit: ask for the plan before any artifact gets built

- Ask the agent for the plan first
- Confirm the plan, then let it write
- PR #52: `proposal-population-plan.md` came before any LaTeX
- Explicit prompt > clever prompt
- Example: thread on `tobias_2025_autonomy` citation → commit `7037de0`

Agents close threads one at a time — consolidating across many is still you

- Agents fix inside one thread
- Compressing dozens of threads → 5 pages is human work
- PR #52 needed a dedicated trim-plan review
- Important things get dropped otherwise
- PI reads, triages, points

Oversight has to scale up as you near the final artifact

- Early: let it sprint
- Late: read every diff
- Build toggles: `make draft` / `make clean-pdf` / `make final`
- 8 rounds of vision-model figure critique
- Then drop the LLM and finish in matplotlib

When the LLM is fighting you, route around it — don't think all-or-nothing

- It's good at shortlists, first drafts, cross-checks
- It's bad at taste and final polish
- Use it where it's good, swap out where it isn't
- Inkscape, PowerPoint, raw matplotlib are fine
- Seriously: drop the chat window

Agent transcripts and PR threads give you a research diary for free

- Every Copilot session = a transcript
- Every PR thread = an audit log
- Repo has a 50,088-line session log
- Future-you (or onboarding) reads it like a journal
- Zero extra writing required

A student told me he did three months of work in a week — because the plan was clear

- Punchline: "three months in a week"
- Why: plan was clear, target was clear
- Tools just removed friction
- Not faster typing — fewer wrong turns
- Caveat: works when you know the target

Perfectionism comes at the end of the artifact, not the beginning

- First 80%: land the structure
- Last 20%: typography, citations, single-word swaps
- Late commit train on the DAG figure = pure polish
- Edison checklist sweeps rounds 6, 7, 8 near submission
- Don't polish what might get cut

Agents surface contacts beautifully — the personalized email is still yours to write

- Agents are great at named-contact shortlists
- Commits `5a13555`, `2d208a8`, `d5b7eca`, `44a4abb`: surfaced + verified contacts
- The letters are still hand-written
- Commits `cda9b3a`, `30febeb`, `8b375f5`: hand-edited partner letters
- Agents find people; you build relationships

If you're starting on this stack today, do these five things

- Pick one real problem and version everything
- Plan-first, then let the agent execute
- Compose tools — don't pick favorites
- Read the diff. Always.
- The transcript is your lab notebook

Thanks — questions?

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- Slides + figures shared after
- Lab repo private; happy to screen-share
- Ask me anything — process or science